

Ge Yan

Email: geyan@u.nus.edu | Tel: +65 90390664 | [Google Scholar](#) | [Website](#) | [LinkedIn](#)

PhD candidate, Electrical Engineering, National University of Singapore, Singapore

RESEARCH INTERESTS

Reconfigurable wireless communication systems; channel knowledge map; wireless sensing; optimization algorithms; electromagnetic theory; computational imaging; machine learning.

EDUCATION

- **2022/08-2026.10 (expected)** **PhD. in Electrical and Computer Engineering**
 - Institution: National University of Singapore, Singapore
 - Advisors: Prof. Rui Zhang, Prof. Xudong Chen
 - Thesis Title: Reconfigurable MIMO for Wireless Communication: Optimization and Practical Channel Knowledge Acquisition
- **2021/01-2022/06** **MSc. in Electrical and Computer Engineering**
 - Institution: National University of Singapore, Singapore
 - GPA: 4.75/5.00
- **2016/09-2020/06** **B.Eng. in Information Science & Engineering**
 - Institution: Chien-Shiung Wu College, Southeast University, Nanjing, China
 - GPA: 3.89/4.00 (Overall Ranking: 1/114)

RESEARCH/PROJECT EXPERIENCE

At Prof. Rui Zhang's Research Group, National University of Singapore, Singapore.

- **[Ongoing] Channel Knowledge Map (CKM) Modelling from An Electromagnetic Perspective (2026/04-present)**
 - Developing an electromagnetic approach to CKM modeling and reconstruction in contrast to conventional ray-tracing-based approaches.
 - Physics-informed machine learning techniques can be applied to solve the inverse problem, i.e., environment inference given the CKM representation.
 - Represent wireless channels in a physically consistent manner without requiring detailed geometric descriptions of the environment.
- **[Ongoing] Slow Movable Antenna (MA)-Aided Wireless Communications Based on Long-Term Channel Statistics (2024/04-present)**
 - Established a statistical-channel-information-based optimization framework for MA-aided wireless communication, reducing the overhead and facilitating practical deployment of MA-aided systems;
 - Resulted in two first-authored journal publications (one under review) and one magazine paper in preparation.
- **Practical Configuration for Intelligent Reflecting Surface (IRS)-Assisted Wireless Systems (2022/10-2025/09)**
 - Developed practical channel acquisition and quasi-static optimization methods for IRS-aided wireless systems, enabling cost-efficient long-term coverage enhancement.

- Utilized channel measurements available from existing protocols without additional signalling, facilitating seamless integration of IRSs into existing networks.

SELECTED PROJECTS & IMPLEMENTATIONS

At Prof. Rui Zhang's Research Group, National University of Singapore, Singapore.

- **Movable Antenna (MA)-Aided Wireless Communications Prototype (2022/10-2025/09)**
 - Developed a prototype with mechanical antenna positioning controls and integrated it with National Instruments software-defined radio platforms.
 - Designed and conducted experiments to validate the performance of MA-aided wireless communication algorithms under practical system conditions.
 - Led algorithm integration, system debugging, and performance evaluation.
- **Intelligent Reflecting Surface (IRS) Channel Acquisition Prototype (2022/10-2025/09)**
 - Collaborated with a remote implementation team to integrate algorithms from my research into a prototype of IRS-aided wireless communication systems.
 - Provided algorithm implementation and technical support, assisted with system debugging, and contributed to performance evaluation and validation.

PROPOSAL DEVELOPMENT EXPERIENCE

At Prof. Rui Zhang's Research Group, National University of Singapore, Singapore.

- Contributed to research grant proposals on MA-aided systems and channel knowledge maps, including system modeling, technical formulation, and experimental designs.

PUBLICATIONS

- **Journals**
 1. **G. Yan**, L. Zhu, W. Ma, and R. Zhang, "Slow Movable Antenna System Design Based on Cell-Specific Long-Term Angular Power Spectrum," 2026. [Online]. Available: <https://arxiv.org/abs/2605.12192>. (submitted to *IEEE Trans. Wireless Commun.*)
 2. **G. Yan**, L. Zhu, and R. Zhang, "Movable Antenna Aided Multiuser Communications: Antenna Position Optimization Based on Statistical Channel Information," in *IEEE Trans. Commun.*, vol. 74, pp. 1793-1810, 2026.
 3. **G. Yan**, L. Zhu, and R. Zhang, "Power Measurement Enabled Channel Autocorrelation Matrix Estimation for IRS-Assisted Wireless Communication," in *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 1832-1848, Mar. 2025.
 4. **G. Yan**, L. Zhu, H. Sun, and R. Zhang, "Wideband Coverage Enhancement for IRS-Aided Wireless Networks Based on Power Measurement," in *IEEE Wireless Commun. Lett.*, vol. 15, pp. 670-674, 2026.
 5. **G. Yan**, L. Zhu, and R. Zhang, "Passive Reflection Optimization for IRS-Aided Multicast Beamforming with Discrete Phase Shifts," in *IEEE Wireless Commun. Lett.*, vol. 12, no. 8, pp. 1424-1428, Aug. 2023.
 6. L. Zhu, H. Mao, **G. Yan**, W. Ma, Z. Xiao, and R. Zhang, "Movable and reconfigurable antennas for 6G: Unlocking electromagnetic-domain design and optimization," *npj Wireless Technol.*, vol. 2, no. 1, pp. 1-10, Feb. 2026.
- **Magazines**

1. H. Sun, L. Zhu, **G. Yan**, and R. Zhang, "Quasi-Static IRS Reflection Design for Network Coverage Enhancement: A Practical RSRP-Based Approach," in *IEEE Wirel. Commun.*, May 2026, early access, DOI: 10.1109/MWC.2026.3692436.
- **Conferences**
 1. **G. Yan** and R. Zhang, "Antenna Position Optimization Based on Statistical Channel Information for Movable Antenna Aided Multiuser Communications," *2025 IEEE Globecom Workshops (GC Wkshps)*, Taipei, Taiwan, 2025, pp. 514-519.
 2. **G. Yan**, L. Zhu, and R. Zhang, "Channel Autocorrelation Estimation for IRS-Aided Wireless Communications Based on Power Measurements," *2023 IEEE Globecom Workshops (GC Wkshps)*, Kuala Lumpur, Malaysia, 2023, pp. 1457-1462.

AWARDS/FELLOWSHIPS

- President's Graduate Fellowship (2022/08-2026/07) National University of Singapore
- National Scholarship (Top 1%, 2018/10) Chinese Ministry of Education
- President Scholarship (Top 1%, 2017/10) Southeast University, Nanjing, China

TEACHING EXPERIENCE

Served as a Teaching Assistant for multiple undergraduate and graduate courses throughout my PhD, all at the National University of Singapore; responsibilities included lecture material preparation, student consultation, assignment/quiz grading, tutorial instruction, and project supervision; skilled in identifying students' misconceptions and providing targeted guidance to facilitate conceptual understanding.

- EE4032 Blockchain Engineering (2022 Fall, 2023 Fall, 2024 Fall)
- CEG5103 Wireless and Sensor Networks for IoT (2023 Spring, 2024 Spring)
- EE5139 Information Theory and Its Applications (2023 Fall)
- CEG5102 Wireless Communications for IoT (2025 Fall)
- EE3131C Communication Systems (2026 Spring)
- EE5131/6131 Wireless Communications (Advanced) (2026 Spring)

PROFESSIONAL SERVICES

- Chaired session 3 of workshop 34 at IEEE Globecom 2025, Taipei, Taiwan.
 - Mon, December 8, 2025, 14:00-15:30 in Room 101B at the Taipei International Convention Center (TICC).
- IEEE Student Member.
- Reviewer for IEEE TWC, TCOM, WCL, etc.

TECHNICAL SKILLS

Skilled in algorithm development and simulation conduction integrated with various physically grounded setups and efficient optimization tools, particularly for signal processing, estimation and optimization problems.

- **Wireless:** MIMO, beamforming, channel models, computational electromagnetics.
- **Algorithms:** optimization, numerical methods, signal processing, inverse problems.
- **Machine learning:** deep learning, computer vision, PyTorch, TensorFlow.

- **Programming & tools:** Python, MATLAB, C/C++, COMSOL, CST, CVX, Git, Linux.

INDUSTRY EXPERIENCE

- **2020/08-2020/12** **AI Research Intern, R&D, SenseTime, Shanghai, China**
 - Participated in image recognition and segmentation projects.
 - Investigated GAN-based video generation.